Notes: Suárez Serrato and Zidar (AER, 2016)

Who Benefits from State Corporate Tax Cuts? A Local Labor Markets Approach with Heterogeneous Firms

Contents

1 Big picture

- **Question.** Who benefits from state corporate tax cuts: workers, landowners, or firm owners? The paper studies the local incidence of state corporate taxation when both firms and workers are imperfectly mobile.
- **Standard view being challenged.** In a highly open economy with mobile capital, the conventional view is that workers bear most or all of the burden of corporate taxes because capital can leave and labor is left with lower demand.
- **Key theoretical idea.** Firms are heterogeneous in their location-specific productivity. Some firms are inframarginal in a location because they are especially productive there, so they do not leave after modest tax changes. This allows firm owners to bear part of the tax burden.
- **Local labor market channel.** A business tax cut lowers firms' tax liability and cost of capital, attracts establishments, increases local labor demand, raises wages, induces migration, and increases housing costs.
- **Empirical setting.** The authors use US state corporate tax changes, state apportionment rules, and matched firm-establishment data to construct local business tax changes for 490 county-groups over decades from 1980 to 2010.
- **Main reduced-form result.** A 1 percent cut in local business taxes raises local establishment growth by roughly 3 to 4 percent over a ten-year period.
- **Main incidence result.** The paper rejects the benchmark in which workers bear 100 percent of the incidence and firm owners bear zero. Baseline estimates put about 40 percent of the burden on firm owners, about 30 to 35 percent on workers, and about 25 to 30 percent on landowners.
- **Policy implication.** Business mobility alone does not imply very low revenue-maximizing state corporate tax rates. However, fiscal externalities through personal income and sales tax bases, plus apportionment rules, are important for revenue calculations.

2 Incremental contribution of the paper

- **From representative firms to heterogeneous firms.** Earlier open-economy corporate tax incidence models often imply that mobile capital escapes the tax and workers bear the burden. This paper adds heterogeneous, location-specific firm productivity, allowing firms to be imperfectly mobile and inframarginal.
- **A bridge between corporate tax incidence and local labor markets.** The paper connects corporate tax incidence to the local labor markets framework used to analyze place-based shocks and policies. It extends that literature by giving the demand side richer microfoundations through firm location choices.
- **A new local business tax measure.** The paper implements state corporate tax rules, including apportionment weights, and uses matched firm-establishment data to measure the effective tax rate businesses face in a local area.
- **Direct mapping from reduced forms to incidence.** Instead of relying only on calibrated structural assumptions, the paper shows how four reduced-form effects - on establishments, population, wages, and rents - identify the incidence shares.

- **Empirical test of the conventional view.** The paper does not simply present an alternative calibration. It statistically rejects the conventional view that workers bear all of the incidence and firm owners bear none.
- **Revenue-maximizing tax analysis.** The paper links incidence to efficiency and state fiscal policy, showing that fiscal externalities and apportionment rules matter for revenue-maximizing corporate tax rates.

3 Data and measurement

3.1 Geography and outcomes

- The main geographic unit is the consistent public-use microdata area, or county-group, which can be followed consistently through Census and ACS data.
- The analysis uses 490 county-groups, observed over decade changes from 1980-1990, 1990-2000, and 2000-2010.
- Establishment counts come from County Business Patterns and are available annually. Population comes from BEA data.
- Wages and housing costs are constructed from Census and ACS microdata and adjusted for changes in worker and housing-unit composition.
- The key outcomes are establishment growth, population growth, adjusted wage growth, and adjusted rental cost growth.

Interpretation: The empirical design is explicitly local. The paper asks how a local area changes relative to other areas when its business tax environment changes.

3.2 State corporate taxes and apportionment rules

- Multi-state firms do not simply pay the statutory tax rate of the state where one establishment is located. States apportion national profits using weighted averages of in-state payroll, property, and sales activity.
- The authors use state tax rates and apportionment weights to construct firm-specific and local effective corporate tax rates.
- The local business tax measure combines taxes on C-corporations with taxes on pass-through businesses that are taxed through the individual income tax system.
- The apportionment formula generates identifying variation because a firm's tax liability depends not only on the local state's tax changes but also on rules and rates in other states.

Interpretation: The tax measure is richer than a statutory state corporate tax rate. It approximates the tax incentives faced by actual establishments operating through multi-state firms.

3.3 Controls and complementary shocks

- The main controls include Bartik labor demand shocks, changes in government spending per capita, changes in state investment tax credits, and changes in other state taxes.

- The paper also uses external tax variation from other states' tax and apportionment rules as a robustness check.
- Population weights are used in the regressions, and standard errors are clustered by state.

Interpretation: These controls are designed to separate business tax variation from contemporaneous policy changes and local economic shocks.

4 Model and empirical specifications

4.1 Households and local labor supply

Households choose where to live based on wages, rents, amenities, and idiosyncratic preferences. A simple indirect utility representation is

$$V_{nc} = a_0 + \ln w_c - \alpha \ln r_c + A_c + \varepsilon_{nc}.$$

- Higher wages attract workers.
- Higher housing costs deter workers.
- Idiosyncratic location preferences make workers imperfectly mobile.
- Housing supply determines how strongly migration raises rents.

Interpretation: Workers can benefit from tax cuts through wages, but part of the wage gain may be capitalized into higher housing costs.

4.2 Firms and local labor demand

Firms are monopolistically competitive and differ in location-specific productivity. They choose location and scale after considering wages, capital costs, and business taxes. In reduced form, local labor demand shifts when business taxes change.

- A tax cut mechanically raises after-tax profits and reduces the effective cost of capital.
- More establishments enter the local market, shifting labor demand out.
- Higher equilibrium wages partially offset the profit gain.
- Firm heterogeneity limits mobility: some firms remain in high-tax locations if the productivity advantage is large enough.

Interpretation: The model's central departure from the standard view is that firm owners are not perfectly mobile at the margin. Location-specific productivity gives them rents that can be taxed.

4.3 Incidence formulae

The paper maps tax changes into welfare changes for three groups:

$$\text{Workers: } \widehat{w} - \alpha\widehat{r}, \qquad \text{Landowners: } \widehat{r}, \qquad \text{Firm owners: } \widehat{\pi}.$$

- Workers care about real disposable income: wages net of housing costs.
- Landowners benefit when rents rise.
- Firm owners benefit when after-tax profits rise.

Interpretation: The incidence of a corporate tax cut is not read off from wages alone. The full local equilibrium also includes rents and profits.

4.4 Reduced-form specification

The main reduced-form regressions relate ten-year local outcome changes to ten-year changes in net-of-business-tax rates:

$$\Delta \ln Y_{c,t} = \beta^Y \Delta \ln(1 - \tau^b_{c,t}) + D'_{s,t}\Psi + u_{c,t}.$$

- $Y_{c,t}$ is establishment counts, population, wages, or rents.
- $\Delta \ln(1 - \tau^b_{c,t})$ is a local business tax cut, measured as an increase in the business tax keep-rate.
- $D_{s,t}$ includes decade indicators and regional controls, with richer specifications adding Bartik shocks and policy controls.

Interpretation: The coefficients $\beta^E, \beta^N, \beta^W$, and β^R are the empirical moments used to infer incidence.

5 Identification and empirical design

5.1 What identifies the reduced-form effects

- The paper exploits within-state changes in corporate tax rates and apportionment rules across decades.
- The long-difference design removes time-invariant county-group characteristics.
- The baseline identification assumption is that, conditional on controls, business tax changes are not driven by unobserved local shocks that would otherwise predict future establishment, population, wage, or rent growth.

Interpretation: The design is not a randomized experiment, but the paper uses multiple robustness exercises to show that the tax variation does not appear to be a response to preexisting local trends.

5.2 External tax variation

- Because multi-state firms are taxed using apportionment rules, a local area's effective business tax rate can change when other states change tax rates or rules.
- These external changes are plausibly less related to local economic conditions.
- The paper finds roughly symmetric effects: local tax cuts raise establishment growth, while external tax cuts in competing locations reduce local establishment growth.

Interpretation: External tax variation is a powerful robustness check because it uses tax changes outside the local area to predict local business incentives.

5.3 Identification of incidence

- The incidence shares are identified by the four reduced-form effects: establishment growth, population growth, wage growth, and rent growth.
- Worker incidence is directly related to $\beta^W - \alpha\beta^R$.
- Landowner incidence is directly related to β^R.
- Firm-owner incidence requires using the model to infer how wage increases reduce after-tax profits.

Interpretation: The incidence exercise is transparent: it does not rely only on a black-box simulation. The key empirical inputs are visible reduced-form moments.

6 Main results

6.1 Effects on local economic activity

- The baseline estimate implies that a 1 percent cut in local business taxes raises establishment growth by about 4.07 percent over ten years.
- The estimate remains between roughly 3.2 and 4.1 percent after controlling for Bartik shocks, government spending, investment tax credits, and external tax changes.
- Population growth responds positively, with estimates around 3.7 to 4.3 percent in the specifications shown.
- Wage and rent responses are positive but less precisely estimated.

Interpretation: Business tax cuts appear to attract establishments and workers. The positive wage and rent estimates are consistent with the local labor demand mechanism, even if they are noisier than the establishment response.

6.2 Reduced-form incidence

- In the baseline reduced-form incidence estimates, firm owners receive about 42 percent of the benefit of a business tax cut, workers about 28 percent, and landowners about 30 percent.
- Across reduced-form specifications, firm-owner shares are consistently far above zero.
- The paper strongly rejects the conventional view that workers bear 100 percent and firm owners bear 0 percent.

Interpretation: The reduced-form incidence estimates are the central empirical rejection of the standard open-economy intuition.

6.3 Structural estimates

- The paper estimates structural parameters that rationalize the reduced-form responses, including labor supply, labor demand, housing supply, and firm productivity dispersion parameters.
- In the baseline structural incidence table, firm owners receive 36.5 percent, workers 22.5 percent, and landowners 41.0 percent.
- Across structural specifications, firm owners receive a substantial share, often around 35 to 55 percent.

Interpretation: The structural estimates corroborate the reduced-form results. They also show that the implied elasticities are broadly plausible relative to existing local labor market estimates.

6.4 Revenue-maximizing tax rates

- If one only considers establishment mobility, revenue-maximizing state corporate tax rates are around 32 percent, much higher than actual state rates.
- Once fiscal externalities on personal income and sales tax bases are included, revenue-maximizing rates are much lower.
- Sales apportionment can reduce the location distortion from corporate taxes because destination-based sales weights are less tied to the location of production.

Interpretation: The paper's efficiency analysis does not imply that states should simply raise corporate taxes to 32 percent. Fiscal externalities and apportionment rules are central for policy design.

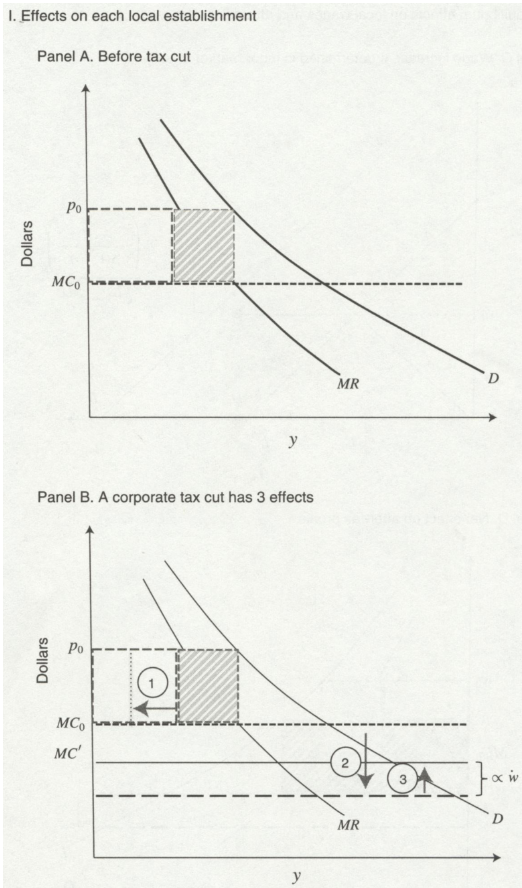
7 Tables and figures

7.1 Figure 1: Mechanism of a corporate tax cut

Read:

- Panels A and B show how a corporate tax cut mechanically changes firm profits: tax liability falls, the capital wedge falls, and firms enter the local area.
- Panels C and D show the general equilibrium: establishment entry raises local labor demand, wages increase, workers move in, rents rise, and higher wages partially compete away firm profits.
- The figure visually organizes the main incidence channels.

Economic message: Corporate tax cuts affect more than firms' statutory tax liabilities. They change local factor prices and distribute benefits across firm owners, workers, and landowners.



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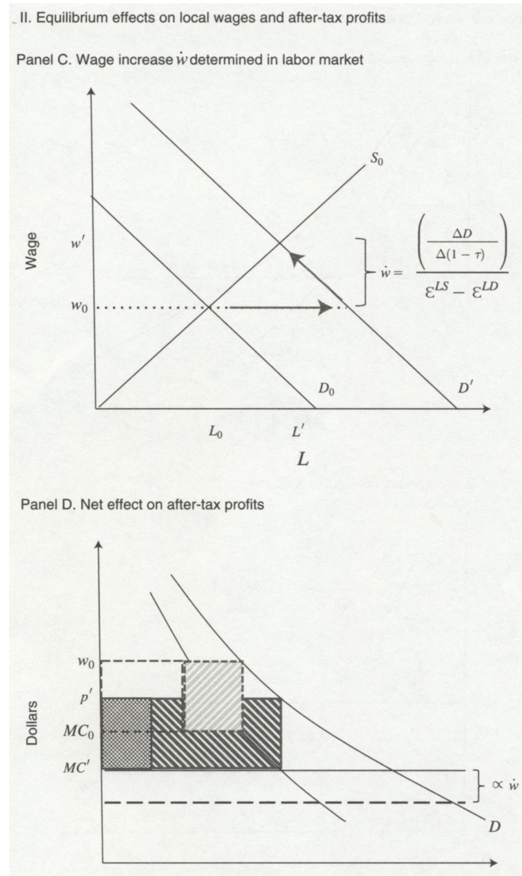


(a) Figure 1, Panels A and B

(b) Figure 1, Panels C and D

Figure 1: The mechanism linking tax cuts to firm profits, wages, and rents.

7.2 Table 1 and Figure 2: Identification and tax variation

Read:

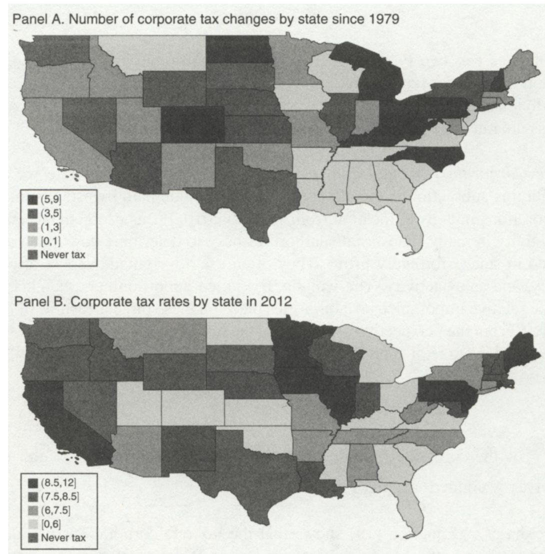
- Table 1 shows the exact mapping from reduced-form effects to local incidence and structural parameters.
- The key reduced-form moments are the effects on wages, population, rents, and establishments.
- Figure 2 documents substantial cross-state variation in corporate tax changes since 1979 and in corporate tax rates in 2012.
- The maps show that many states changed corporate tax rates, while some states never had a corporate income tax.

Economic message: The empirical strategy relies on meaningful geographic and time-series variation in business taxes, while the theory explains how those reduced-form responses translate into welfare incidence.

TABLE 1—IDENTIFICATION OF LOCAL INCIDENCE ON WELFARE AND STRUCTURAL PARAMETERS

Stakeholder (benefit)	Incidence	Identified by	
Panel A. Local incidence			
Workers (disposable income)	$\dot{w} - \alpha \dot{r}$	$\beta^W - \alpha\beta^R$	
Landowners (housing costs)	$\dot{r}$	β^R	
Firm owners (after-tax profit)	$1 + \gamma(\varepsilon^{PD} + 1)(\dot{w}_c - \frac{\delta}{\gamma})$	$1 + \left(\frac{\beta^N - \beta^E}{\beta^W} + 1\right)\left(\beta^W - \frac{\delta}{\gamma}\right)$	
Panel B. Structural parameters			
Worker mobility	Firm mobility	Housing supply	Product demand
$\sigma_W = \frac{\beta^W - \alpha\beta^R}{\beta^N}$	$\sigma_F = \frac{\gamma\beta^W}{\beta^E}\left(\frac{1}{\beta^E - \beta^N - \beta^W} - 1\right)$	$\eta = \frac{\beta^N + \beta^W}{\beta^R} - 1$	$\varepsilon^{PD} = \frac{\beta^N + \beta^W - \beta^E}{\gamma\beta^W}$

(a) Table 1: Incidence identification



(b) Figure 2: State corporate tax variation

Table 1 and Figure 2: Reduced-form moments and the tax variation that identifies them.

7.3 Table 2 and Figure 3: Data and apportionment rules

Read:

- Table 2 reports summary statistics for outcomes, tax variables, apportionment weights, Bartik shocks, and decade changes.
- The decadal data show large variation in establishment growth, population growth, wages, rents, and tax changes.
- Figure 3 shows that states shifted increasingly toward sales-based apportionment over time.
- In 1980, many states used equal one-third weights; by 2010, many states placed much more weight on sales.

Economic message: Apportionment rules are not a technical detail. They determine how strongly a firm's tax liability depends on where it places workers, property, and sales.

Table 2—Summary Statistics

Variable	Mean	SD	Min	Max	Observations
Annual outcome data from BEA and CBP					
Year	1995	8.9	1980	2010	15,190
log population: $\ln N_{c,t}$	13.8	1.1	10.9	16.1	15,190
log employment: $\ln L_{c,t}$	13.2	1.2	9.4	15.6	15,190
log establishments: $\ln E_{c,t}$	10.0	1.2	6.5	12.4	15,190
Annual data on apportionment rules and corporate, personal, and business tax rates					
State corporate tax apportionment parameters					
Payroll apportionment weight: $\theta^{w}_{s,t}$	22.7	11.6	0.0	33.3	15,190
Property apportionment weight: $\theta^{\rho}_{s,t}$	22.8	11.6	0.0	33.3	15,190
Sales apportionment weight: $\theta^{x}_{s,t}$	54.5	23.2	25	100	15,190
Corporate income					
Rate: $\tau^{c}_{s,t}$	6.6	3.0	0.0	12.3	15,190
Percent change in net-of-rate: $\Delta\ln(1-\tau^{c})_{s,t,t-1}$	−0.01	0.4	−5.4	3.8	15,190
Personal income					
Effective rate: $\tau^{i}_{s,t}$	2.6	1.7	0.0	7.4	15,190
Percent change in net-of-rate: $\Delta\ln(1-\tau^{i})_{s,t,t-1}$	0.03	0.2	−3.3	2.5	15,190
Business income					
Rate: $\tau^{b}_{c,t}$	3.1	1.1	0.3	5.4	15190
Percent change in net-of-rate: $\Delta\ln(1-\tau^{b})_{c,t,t-1}$	−0.01	0.2	−1.8	1.2	15,190
Year	2000	8.2	1990	2010	1,470
Decadal data					
Percentage change in:					
Population: $\Delta\ln N_{c,t,t-10}$	11.2	10.4	−16.6	76.1	1,470
Establishments: $\Delta\ln E_{c,t,t-10}$	15.2	16.5	−23	126.2	1,470
Adjusted wages: $\Delta\ln w_{c,t,t-10}$	−2.8	7.2	−31.2	14.9	1,470
Adjusted rents: $\Delta\ln r_{c,t,t-10}$	8.5	12.0	−41.4	43.4	1,470
Net-of-corp.-rate: $\Delta\ln(1-\tau^{c})_{s,t,t-10}$	−0.1	1.1	−5.4	4.5	1,470
Net-of-pers.-rate: $\Delta\ln(1-\tau^{i})_{s,t,t-10}$	−1.3	1.1	−5.3	1.3	1,470
Net-of-bus.-rate: $\Delta\ln(1-\tau^{b})_{c,t,t-10}$	−0.8	0.6	−2.8	1.3	1,470
Gov. expend./capita: $\Delta\ln G_{c,t,t-10}$	0.0	0.6	−13.3	11.6	1,470

(a) Table 2: Summary statistics

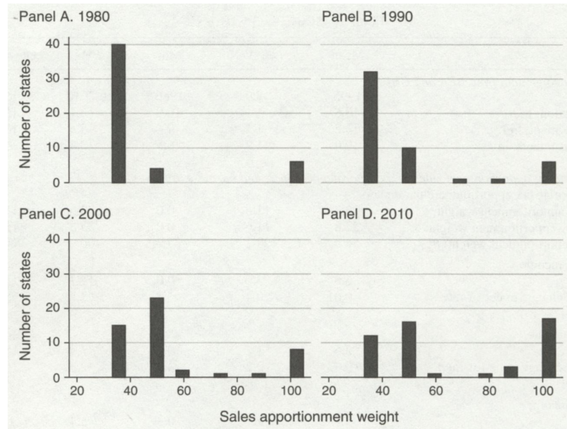


Figure 3. Histogram of Sales Apportionment Weights by Decade

:: This figure shows a histogram of the weight on sales activity that states use to apportion the national

(b) Figure 3: Sales apportionment weights

Table 2 and Figure 3: Outcome variation and the evolution of corporate tax apportionment rules.

7.4 Table 3 and Table 4: Calibration and reduced-form effects

Read:

- Table 3 lists calibrated parameters used in incidence formulae: the output elasticity ratio, housing cost share, labor output elasticity, and product demand elasticity.
- Baseline values are shown in bold in the original table.
- Table 4 reports the core reduced-form effects over ten years.
- The establishment-growth coefficient is about 4.07 in the baseline and remains positive under several controls.

Economic message: The empirical effect of business tax cuts on establishment growth is the main reduced-form force behind the incidence estimates. Calibration choices matter, but the main conclusion is robust across reasonable values.

Table 3—Calibrated Parameters used in Incidence Formulae

Parameter	Values	Sources
Parameters for reduced-form implementation		
Ratio of elasticities: γ/δ	{**0.90**, 0.50, 0.75}	**BEA**
Housing cost share: α	{**0.30**, 0.50, 0.65}	**CEX**, Albouy (2008), Moretti (2013)
Additional parameters for structural implementation		
Output elasticity of labor: γ	{**0.15**, 0.20, 0.25}	**IRS**, BEA, Kline and Moretti (2014a)
Elasticity of product demand: ε^{PD}	{**−2.5**, −3.5, estimated}	Between **Head and Mayer (2014)** and ε^{LD} in **Hamermesh (1993)**

(a) Table 3: Calibrated parameters

Table 4—Effects of Business Tax Cuts on Local Economic Activity over 10 Years

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Establishment growth						
Δln net-of-business-tax rate	4.07	3.35	4.06	4.14	3.91	3.24
	(1.82)	(1.43)	(1.83)	(1.80)	(1.78)	(1.41)
Bartik		0.59				0.57
		(0.19)				(0.18)
Δln gov expend/capita			−0.01			−0.01
			(0.01)			(0.01)
Δ State ITC				−0.46		−0.17
				(0.32)		(0.30)
Change in other states' taxes					−4.66	−4.18
					(1.60)	(1.43)
Observations	1,470	1,470	1,470	1,470	1,470	1,470
R^2	0.472	0.491	0.472	0.475	0.481	0.500
	Population growth		Wage growth		Rental cost growth	
	(1)	(2)	(1)	(2)	(1)	(2)
Panel B. Other outcomes						
Δln net-of-business-tax rate	4.28	3.74	1.45	0.78	1.17	0.32
	(1.65)	(1.48)	(0.94)	(0.82)	(1.44)	(1.37)
Bartik		0.44		0.56		0.70
		(0.19)		(0.08)		(0.27)
Observations	1,470	1,470	1,470	1,470	1,470	1,470

(b) Table 4: Reduced-form effects

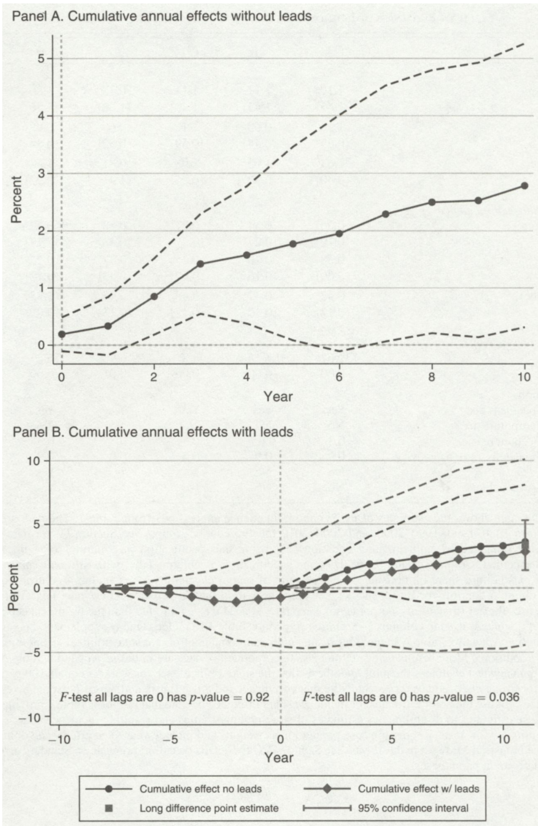
Table 3 and Table 4: Parameters and local economic responses to business tax cuts.

7.5 Figure 4 and Table 5: Dynamics and reduced-form incidence

Read:

- Figure 4 shows cumulative annual effects of business tax cuts on establishment growth.
- The dynamic evidence shows little evidence of pre-trends and a gradual post-cut increase in establishments.
- Table 5 implements the reduced-form incidence formulae.
- In the baseline column, the shares are roughly 30 percent for landowners, 28 percent for workers, and 42 percent for firm owners.

Economic message: The timing evidence supports the identifying assumption, and the reduced-form incidence calculations show that firm owners receive a large share of the benefit from corporate tax cuts.



(a) Figure 4: Dynamic establishment response

TABLE 5—ESTIMATES OF ECONOMIC INCIDENCE USING REDUCED-FORM EFFECTS

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Incidence						
Landowners	1.17	1.17	1.17	0.32	1.86	0.62
	(1.43)	(1.43)	(1.43)	(1.36)	(1.56)	(0.60)
Workers	1.1	0.69	1.1	0.68	0.98	0.58
	(0.59)	(0.44)	(0.59)	(0.52)	(0.84)	(0.33)
Firm owners	1.63	1.63	2.08	0.81	1.54	0.9
	(0.90)	(0.90)	(0.95)	(1.4)	(0.92)	(0.34)
Panel B. Share of incidence						
Landowners	0.30	0.34	0.27	0.18	0.42	0.29
	(0.19)	(0.24)	(0.2)	(0.48)	(0.17)	(0.16)
Workers	0.28	0.20	0.25	0.37	0.22	0.28
	(0.09)	(0.16)	(0.07)	(0.43)	(0.12)	(0.08)
Firm owners	0.42	0.47	0.48	0.45	0.35	0.43
	(0.12)	(0.10)	(0.17)	(0.13)	(0.09)	(0.10)
Conventional view test						
χ^2 of ($S^W = 1, S^F = 0$)	132.67	108.14	48.8	6.96	76.27	195.92
p-value	0.00	0.00	0.00	0.01	0.00	0.00
Specification						
Net-of-business tax	Yes	Yes	Yes	Yes	Yes	No
Net-of-corporate tax	No	No	No	No	No	Yes
Housing share α	0.3	0.65	0.3	0.3	0.3	0.3
Output elasticity ratio δ/γ	0.9	0.9	0.5	0.9	0.9	0.9
Bartik	No	No	No	Yes	Yes	No

(b) Table 5: Reduced-form incidence

Figure 4 and Table 5: Dynamic validation and reduced-form incidence estimates.

7.6 Table 6 and Table 7: Structural parameters and structural incidence

Read:

- Table 6 estimates structural parameters using minimum distance methods.
- The estimated parameters include idiosyncratic firm productivity dispersion, worker preference dispersion, and housing supply elasticity.
- Table 7 reports structural incidence results under several parameter assumptions.
- The baseline structural shares are approximately 41 percent for landowners, 22.5 percent for workers, and 36.5 percent for firm owners.

Economic message: The structural estimates support the reduced-form conclusion: firm owners bear a substantial share of the incidence because firms are not perfectly mobile.

TABLE 6—MINIMUM DISTANCE ESTIMATES OF STRUCTURAL PARAMETERS

	Panel A. All shocks						
Calibrated parameters	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Output elasticity γ	0.150	0.150	0.150	0.200	0.250	0.150	0.250
Housing share α	0.300	0.500	0.650	0.300	0.300	0.300	0.500
Elasticity of product demand ε^{PD}	−2.500	−2.500	−2.500	−2.500	−2.500	−4.000	−4.000
Estimated parameters							
Idiosyncratic location productivity dispersion σ^F	0.277	0.271	0.233	0.321	0.304	0.149	0.136
	(0.138)	(0.120)	(0.092)	(0.186)	(0.186)	(0.096)	(0.093)
Idiosyncratic location preference dispersion σ^W	0.829	0.686	0.621	0.845	0.843	0.839	0.649
	(0.282)	(0.260)	(0.230)	(0.294)	(0.295)	(0.294)	(0.253)
Elasticity of housing supply η	0.513	2.185	1.157	1.600	0.707	1.995	2.812
	(1.417)	(6.206)	(2.661)	(5.065)	(2.301)	(7.320)	(13.688)
Overid test (p-value)	0.458	0.390	0.393	0.385	0.444	0.390	0.507

	Panel B. Business tax shock				Panel C. All shocks, estimated ε^{PD}		
Calibrated parameters	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Output elasticity γ	0.150	0.150	0.250	0.150	0.150	0.150	0.250
Housing share α	0.300	0.650	0.300	0.300	0.300	0.650	0.300
Elasticity of product demand ε^{PD}	−2.500	−2.500	−2.500	−4.000	Estimated below		
Estimated parameters							
Idiosyncratic location productivity dispersion σ^F	0.119	0.117	0.106	0.048	0.109	0.105	0.138
	(0.065)	(0.064)	(0.075)	(0.039)	(0.392)	(0.194)	(0.411)
Idiosyncratic location preference dispersion σ^W	0.188	0.128	0.171	0.170	0.892	0.571	0.753
	(0.184)	(0.147)	(0.176)	(0.175)	(0.337)	(0.234)	(0.245)
Elasticity of housing supply η	6.367	5.724	7.328	6.424	1.925	1.783	3.056
	(15.899)	(13.090)	(20.574)	(16.136)	(8.085)	(6.503)	(25.617)
Elasticity of product demand ε^{PD}					−4.704	−4.439	−4.986

(a) Table 6: Structural parameters

TABLE 7—ESTIMATES OF ECONOMIC INCIDENCE USING ESTIMATED STRUCTURAL PARAMETERS

	All shocks			Business tax	All shocks est. ε^{PD}
	(1)	(2)	(3)	(4)	(5)
Panel A. Incidence					
Calibrated parameters					
Output elasticity γ	0.150	0.150	0.150	0.150	0.150
Housing share α	0.300	0.650	0.300	0.300	0.300
Elasticity of product demand ε^{PD}	−2.500	−2.500	−4.000	−2.500	−4.704
					(11.945)
Estimated incidence					
Wages $\dot{w}$	0.944	1.088	0.655	0.839	0.646
	(0.408)	(0.457)	(0.348)	(0.847)	(1.028)
Landowners $\dot{r}$	1.111	0.886	0.428	0.591	0.420
	(1.119)	(1.052)	(1.079)	(1.373)	(1.517)
Workers $\dot{w} - \alpha\dot{r}$	0.611	0.512	0.527	0.662	0.520
	(0.293)	(0.355)	(0.269)	(0.517)	(0.703)
Firm owners $\dot{\pi}$	0.990	0.958	1.110	1.014	1.141
	(0.092)	(0.103)	(0.157)	(0.191)	(1.012)
Elasticity of labor supply ε^{LS}	0.780	0.757	0.958	4.188	0.902
	(0.386)	(0.729)	(0.588)	(4.795)	(0.645)
Elasticity of labor demand ε^{LD}	−1.766	−1.867	−2.457	−2.485	−2.933
	(0.269)	(0.252)	(0.646)	(0.692)	(6.731)
Panel B. Shares of incidence					
Calibrated parameters					
Output elasticity γ	0.150	0.150	0.150	0.150	0.150
Housing share α	0.300	0.650	0.300	0.300	0.300
Elasticity of product demand ε^{PD}	−2.500	−2.500	−4.000	−2.500	−4.704
					(11.945)
Estimated incidence					
Landowners $\dot{r}$	0.410	0.376	0.207	0.261	0.202
	(0.263)	(0.339)	(0.434)	(0.430)	(0.621)
Workers $\dot{w} - \alpha\dot{r}$	0.225	0.217	0.255	0.292	0.250
	(0.134)	(0.197)	(0.185)	(0.142)	(0.290)

(b) Table 7: Structural incidence

Table 6 and Table 7: Structural parameter estimates and implied incidence shares.

7.7 Figure 5 and Table 8: Robustness and revenue-maximizing rates

Read:

- Figure 5 shows that the firm-owner share remains substantial across a range of calibrated product-demand, labor-output, and housing-share values.
- The baseline parameters are conservative in the sense that many alternative calibrations raise the firm-owner share.
- Table 8 reports revenue-maximizing corporate tax rates for selected states.
- Establishment mobility alone implies high revenue-maximizing rates, but fiscal externalities and apportionment considerations bring rates closer to actual statutory rates.

Economic message: The incidence result is not driven by a knife-edge calibration. For policy, the design of the apportionment formula and the fiscal base of the state are crucial.

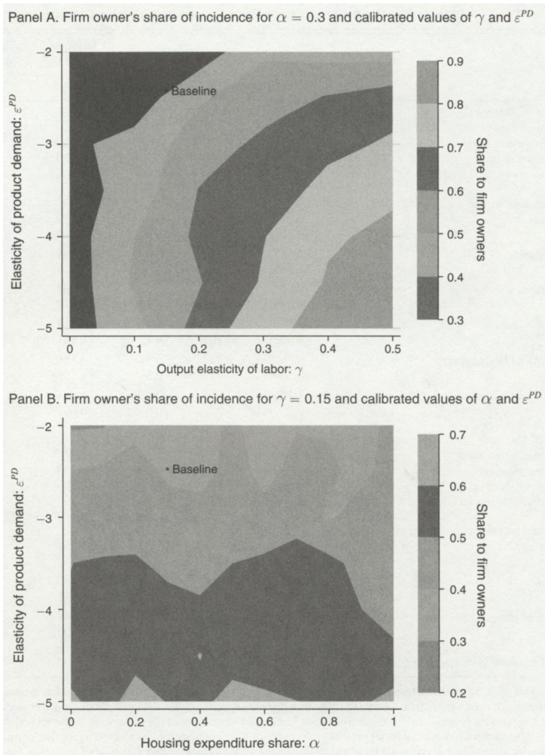


FIGURE 5. ROBUSTNESS OF ECONOMIC INCIDENCE

(a) Figure 5: Robustness of incidence

TABLE 8—REVENUE-MAXIMIZING CORPORATE TAX RATES FOR SELECTED STATES

State	Establishment share E_s	Revenue ratio rev_s^{pers}/rev_s^C	Sales apport. weight θ_s^x	Corporate tax rate τ_s	Revenue max. corp. rate		
					τ_s^*	τ_s^{**}	$\tau_s^{**}/(1-\theta_s^x)$
Kansas	1.0	16.0	33	7.1	30.6	2.2	3.4
New Mexico	0.6	26.1	33	7.6	32.0	1.4	2.1
California	11.7	9.2	50	8.8	32.0	3.7	7.4
Virginia	1.5	18.4	50	6.0	30.1	2.0	3.9
Arizona	1.8	22.1	80	7.0	30.0	1.7	8.3
Indiana	2.0	20.7	90	8.5	32.9	1.8	17.7
Texas	7.2		100	0.0	30.3		
US state average	2.0	21.7	66.1	6.7	31.9	2.8	7.1
US state median	1.4	17.1	50.0	7.1	31.5	2.1	4.4
US state min	0.3	0.4	33.3	0.0	28.6	0.3	0.7
US state max	11.7	141.5	100.0	12.0	36.8	24.1	36.1

(b) Table 8: Revenue-maximizing rates

Figure 5 and Table 8: Robustness of incidence and implications for state tax policy.

8 Short summary

- The paper asks who benefits from state corporate tax cuts in local labor markets.
- It develops a spatial equilibrium model with imperfectly mobile workers and heterogeneous, imperfectly mobile firms.
- Firm heterogeneity creates inframarginal firms that remain in productive locations even when tax rates change.
- The authors construct local effective business tax rates using state tax rates, apportionment rules, and matched firm-establishment data.
- A 1 percent local business tax cut raises establishment growth by about 3 to 4 percent over ten years.
- Worker, rent, population, and establishment responses are mapped into incidence shares.
- Reduced-form estimates put roughly 42 percent of the benefit of a tax cut on firm owners, 28 percent on workers, and 30 percent on landowners in the baseline.
- Structural estimates also imply substantial firm-owner incidence and reject the conventional view that workers bear everything.

- Revenue-maximizing rates depend strongly on fiscal externalities and apportionment rules, not just firm mobility.

9 Conclusion

9.1 Core conclusion

Suárez Serrato and Zidar show that state corporate tax incidence is shared among firm owners, workers, and landowners. The paper's central conclusion is that firm owners bear a substantial share of state corporate taxes because heterogeneous location-specific productivity makes many firms inframarginal in their location choices.

9.2 Economic intuition

The intuition is simple. If a firm is especially productive in a location, a modest state tax increase may reduce profits but not induce relocation. In that case, firm owners cannot fully escape the tax. At the same time, tax cuts attract additional establishments, raising labor demand, wages, population, and rents. Workers benefit through higher real wages, landowners benefit through higher rents, and firm owners benefit through higher after-tax profits.

9.3 Incremental contribution revisited

The paper advances the corporate tax incidence literature by moving beyond the stark open-economy model in which capital is perfectly mobile and workers bear the entire burden. It also advances the local labor markets literature by giving firms a richer role: firm owners can be inframarginal and can bear incidence, not merely shift labor demand.

9.4 Policy implications

The results caution against saying that corporate tax cuts simply help workers or simply help firms. The benefits are distributed. The results also show that state tax policy cannot be evaluated by corporate rates alone: apportionment rules, business composition, housing markets, and fiscal externalities all matter.

9.5 Limitations

The analysis abstracts from transition dynamics, unemployment, endogenous agglomeration, commercial real estate interactions, and heterogeneous incidence across local areas. The empirical design is based on policy variation rather than random assignment, so the robustness tests are essential for the credibility of the results. Housing rents are also an imperfect proxy for landowner welfare.

9.6 Takeaway

The enduring takeaway is that corporate taxes in open local economies need not fall entirely on workers. Once firms are heterogeneous and locally productive, firm owners can bear a meaningful share of the incidence. This changes both the equity interpretation of corporate taxation and the efficiency logic behind state tax competition.